ELEG 4603/5173L Digital Signal Processing
Course Syllabus

General Information:

Instructor: Jingxian Wu — Office: Bell 3181
Email: wuj@uark.edu — Phone: (479) 575-6584
Office Hour: M. F. 1:30-2:30
Lecture location: SCEN 0405 — Lecture: Mo. 2:30-4:30
Lab Location: Bell 3139 — Lab: Fr. 2:30-5:20

Required Material:

- R. Chassaing and D. Reay, *Digital Signal Processing and Applications with the TMS320C6713 and TMS320C6416 DSK*, 2nd Ed., Wiley-IEEE, 2008.
- Luis F. Chaparro, *Signals and Systems Using Matlab*, Academic Press, 2010.
- Software 1: TI Code Composer Studio 3.1
- Software 2: Matlab

Reference:

- S.S. Soliman, M.D. Srinath, *Continuous and discrete signals and systems*, 2nd Ed., Prentice Hall, 1998. (Optional)

Prerequisites:

ELEG 3124 Signals and Systems

- Familiarity with C Programming, Matlab Programming, and Assembly Programming
- Knowledge of continuous-time signals and systems
- Knowledge of convolution, Fourier transform, and Laplace transform
- Knowledge of Memory and Peripherals (e.g., RAM, ROM, ADC, DAC)

Learning Objectives:

To understand the concepts of discrete-time signals and systems, linear system analysis, frequency domain analysis, z-domain analysis. To understand the structure and concept of DSP, to be able to program DSP with assembly language and C, to be able to design various filters, communication applications, and multimedia applications with DSP.

Grading:

- Test 1 22%
- Test 2 22%
- Test 3 22%
- Homework 14%
- Lab 20%

- A: $90 \leq \text{grade} \leq 100$
- B: $80 \leq \text{grade} < 90$
- C: $70 \leq \text{grade} < 80$
- D: $60 \leq \text{grade} < 70$
- F: $0 \leq \text{grade} < 60$

- Each student has 1 personal day: you can miss 1 class without losing point.
- Each missing class will result in a 1 point deduction of your final grade.
- Due dates for homework will be strictly enforced. Late submission within one week after due date will receive a 20% grade deduction, and no credit if submitted after one week from the due date.
- If for some legitimate reason (sickness, death in the family, etc.), you cannot take a **test** on the scheduled day, you must notify the instructor **prior** to the exam.

Online Resources:

- Course materials (Slides, Homework, Labs, References, etc) can be found at http://comp.uark.edu/~wuj/teaching/eleg5173L/eleg5173L.html
- Please check course website **at least once per week** for updates.

Academic

Each University of Arkansas student is required to be familiar with and abide by

Honesty: the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/ Students with questions about how these policies apply to a particular course or assignment should immediately contact their instructor.

Tentative Schedule:

- Week 1 (1/14): Ch.1 Discrete-Time Signals and Systems
- Week 2 (1/21): No class
- Week 3 (1/28): Ch. 1 Discrete-Time Signals and Systems
- Week 4 (2/4): Ch. 1 Discrete-Time Signals and Systems
- Week 5 (2/11): Ch. 2 The Z-Transform
- Week 6 (2/18): Ch. 2 The Z-Transform
- Week 7 (2/25): Ch. 2 The Z-Transform (**Test 1on 3/1 Fr.**)
- Week 8 (3/4): Ch. 3 Discrete-Time Fourier Transform
- Week 9 (3/11): Ch. 3 Discrete-Time Fourier Transform
- Week 10 (3/18): **Spring Break**
- Week 11 (3/25): Ch. 3 Discrete-Time Fourier Transform
- Week 12 (4/1): Ch. 4 Discrete Fourier Transform
- Week 13 (4/12 Fr): Ch. 4 Discrete Fourier Transform (**Test 2 on 4/8 Mo.)**
- Week 14 (4/15): Ch. 4 Discrete Fourier Transform
- Week 15 (4/22): Ch. 5 Discrete-Time Filters
- Week 16 (4/29): Ch. 5 Discrete-Time Filters (dead day 5/3)
- Week 17 (5/7, Tue.): **Test 3 on 5/7 (10:15 AM -12:15 PM, BELL 2284)**

The above schedule is subject to change without prior notice.